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(54) **INFORMATION PROCESSING DEVICE**

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(57) **ABSTRACT**

The information processing device includes a control unit that causes a display unit to display an image indicating the host vehicle and the surroundings when viewed from a virtual viewpoint on the basis of surroundings information regarding surroundings of the host vehicle, and causes the display unit to display an image indicating the registered vehicle in a manner different from an image indicating another vehicle different from the registered vehicle when the registered vehicle is detected.

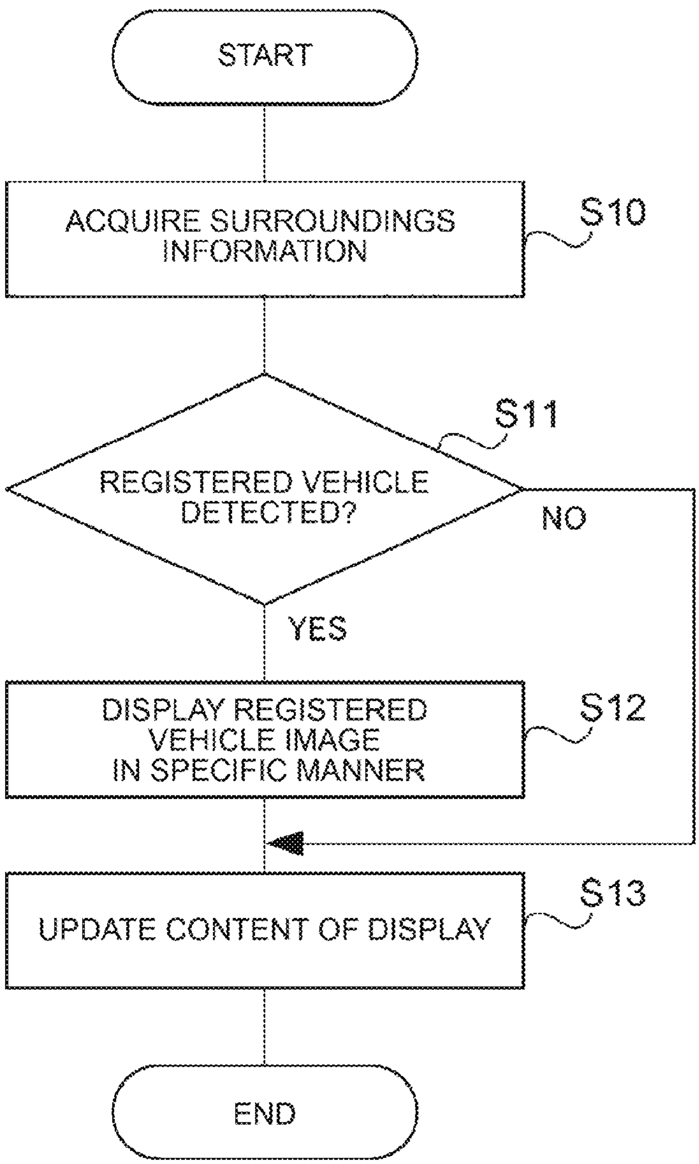


FIG. 1

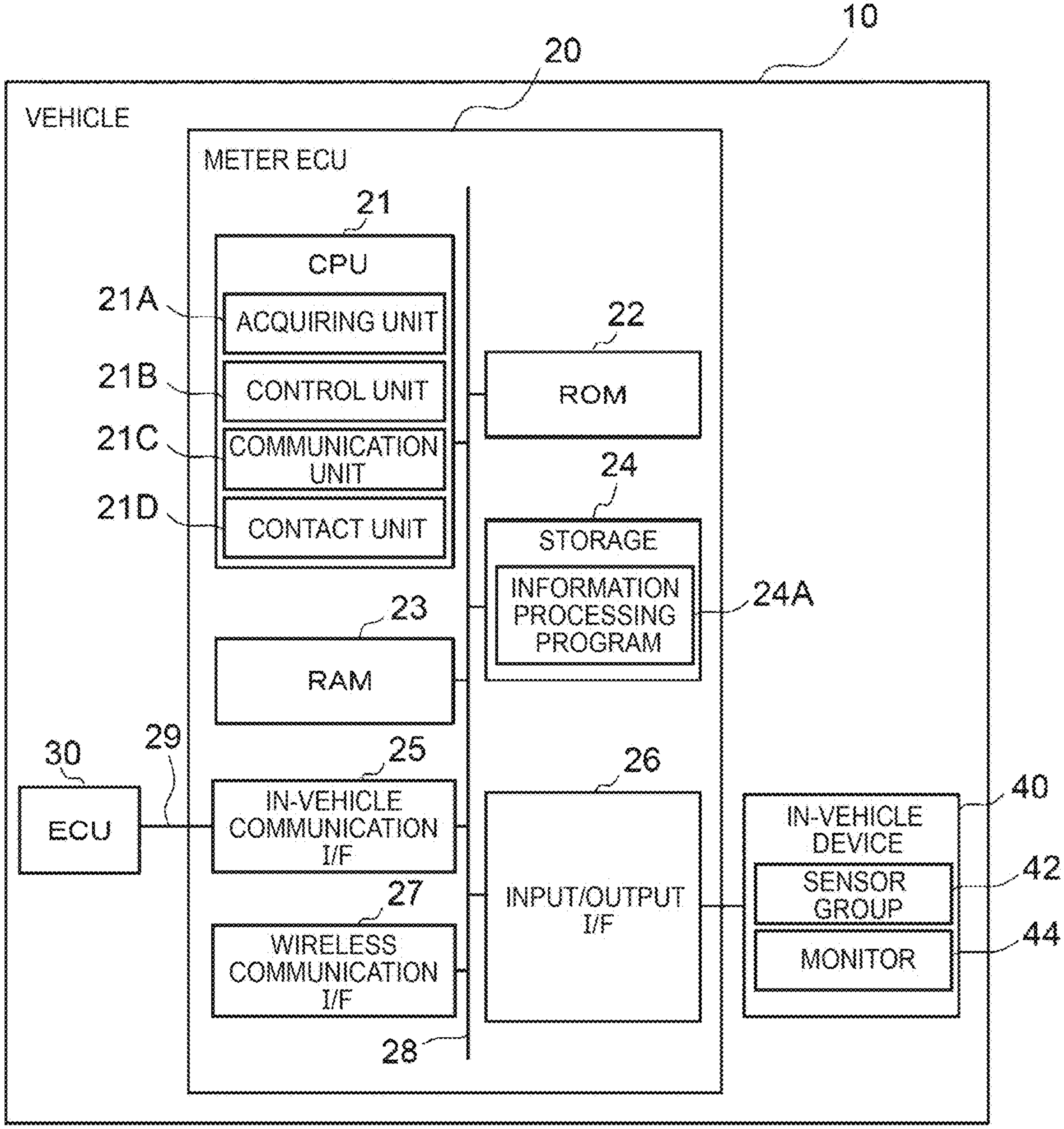

10
20
VEHICLE
METER ECU
21
CPU
21A
ACQUIRING UNIT
21B
CONTROL UNIT
21C
COMMUNICATION UNIT
21D
CONTACT UNIT
22
ROM
24
STORAGE
INFORMATION PROCESSING PROGRAM
24A
23
RAM
30
ECU
29
25
IN-VEHICLE COMMUNICATION I/F
26
INPUT/OUTPUT I/F
40
IN-VEHICLE DEVICE
42
SENSOR GROUP
44
MONITOR
27
WIRELESS COMMUNICATION I/F
28

FIG. 2

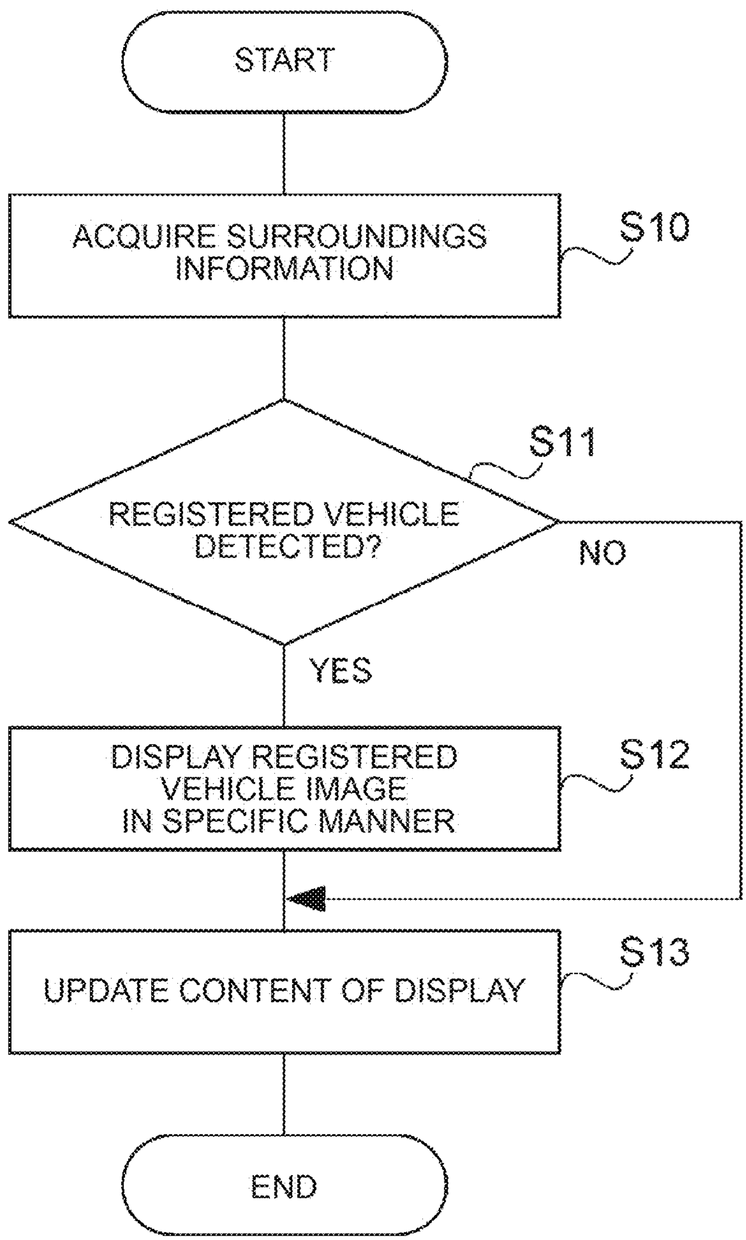

START
ACQUIRE SURROUNDINGS INFORMATION
S10
REGISTERED VEHICLE DETECTED?
S11
NO
YES
DISPLAY REGISTERED VEHICLE IMAGE IN SPECIFIC MANNER
S12
UPDATE CONTENT OF DISPLAY
S13
END

FIG. 3

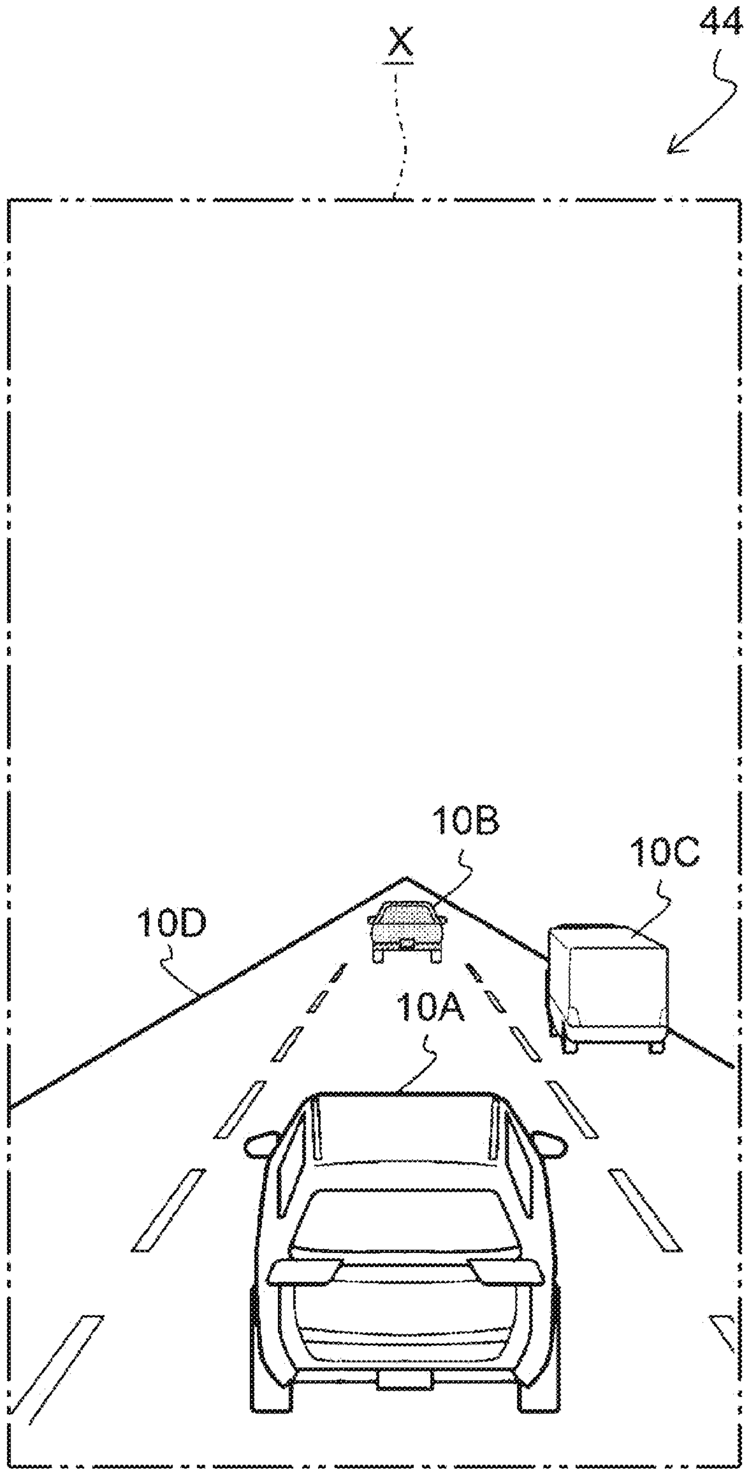


FIG. 4

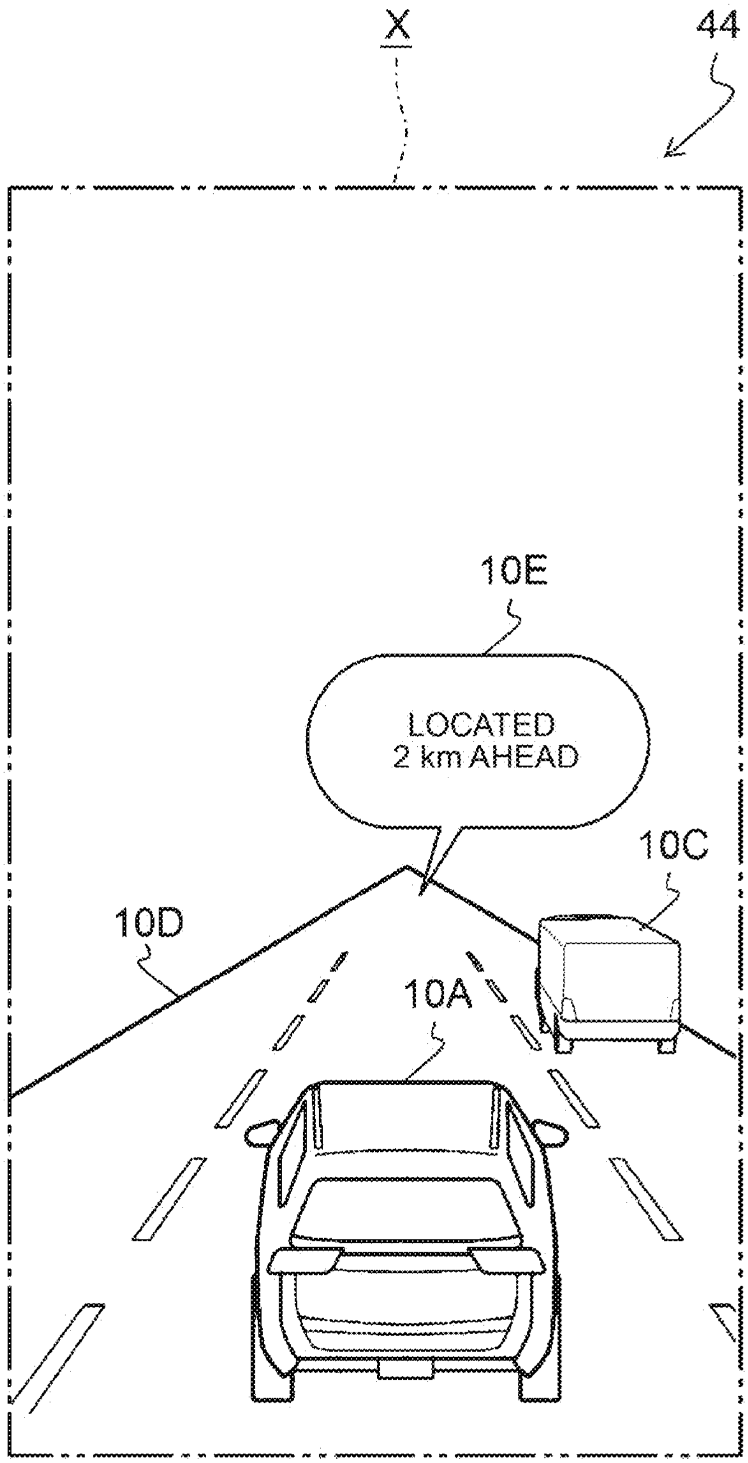


FIG. 5

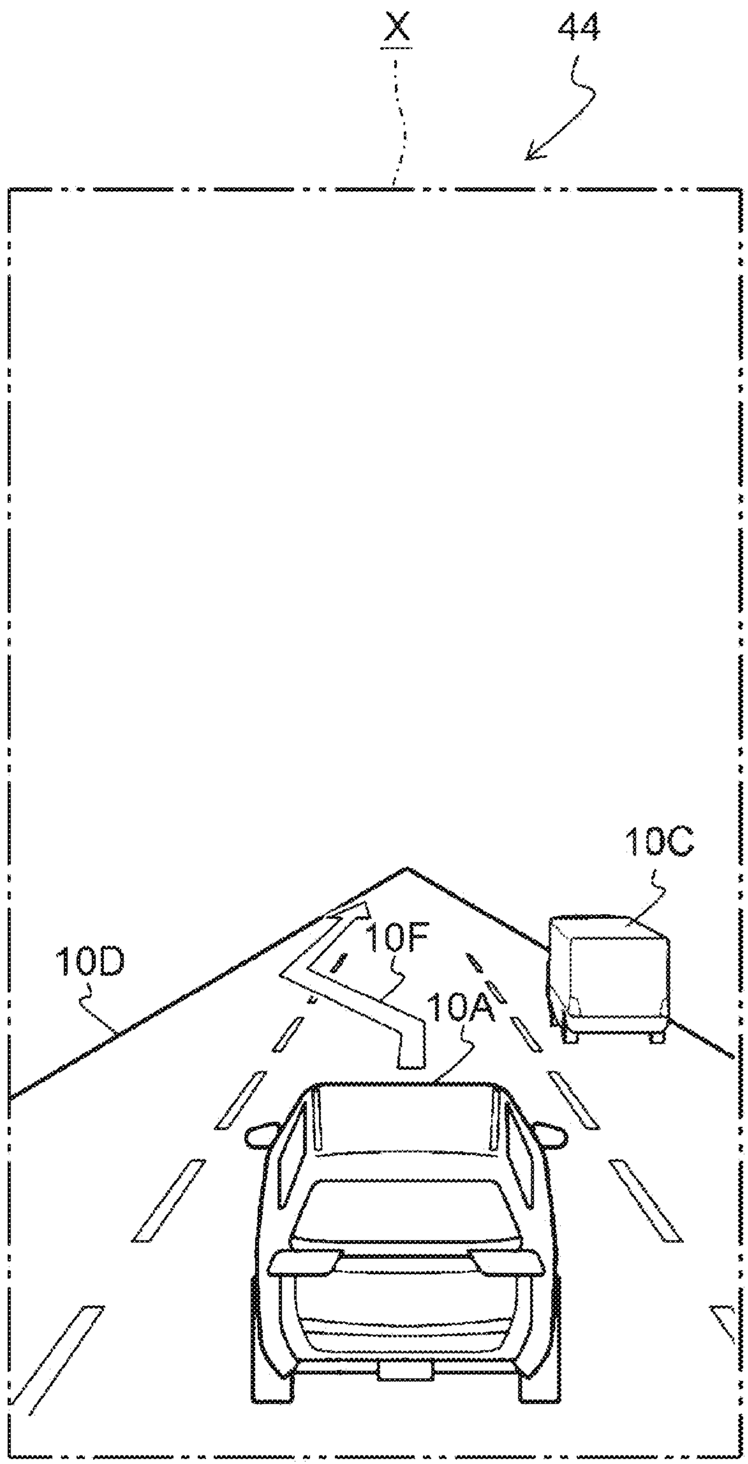
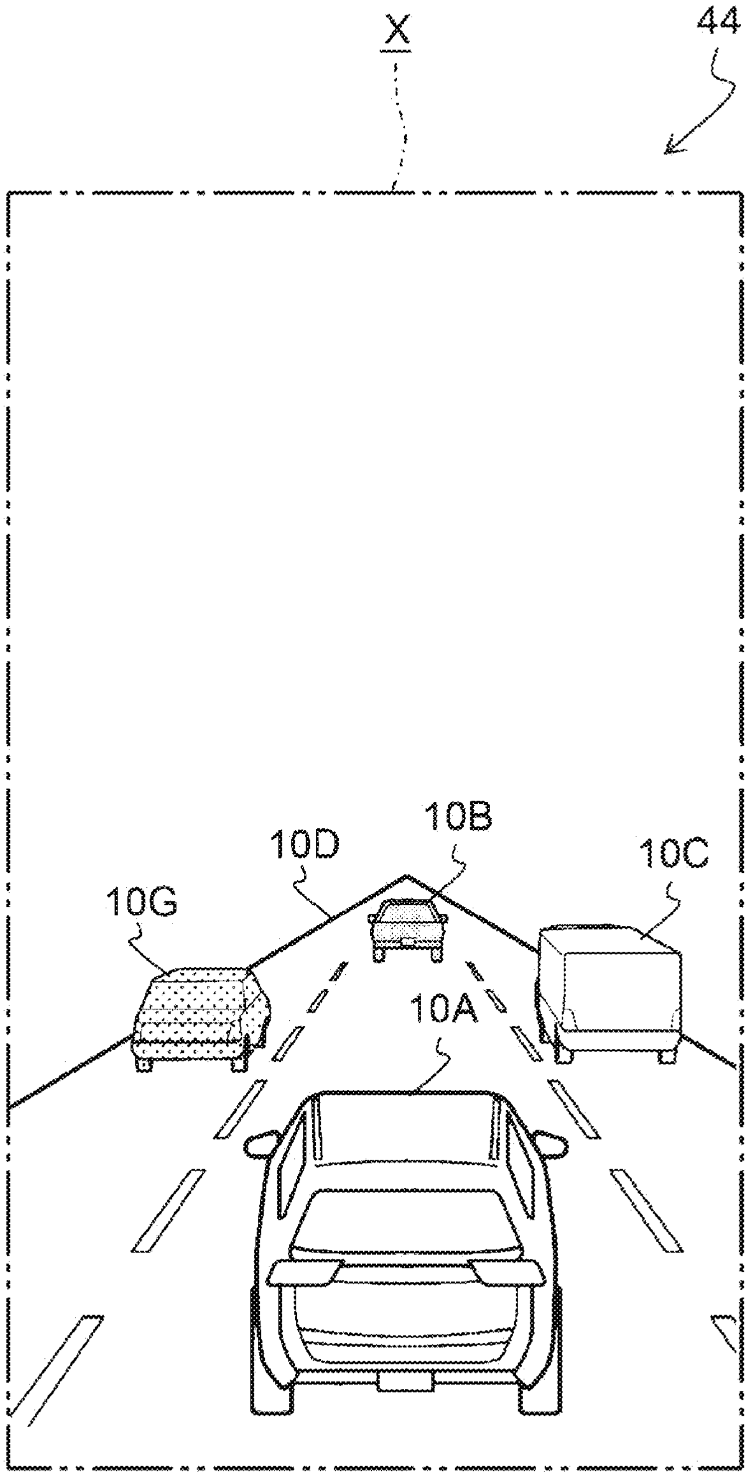


FIG. 6



INFORMATION PROCESSING DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-034291 filed on Mar. 6, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to information processing devices.

2. Description of Related Art

[0003] Japanese Patent No. 7048398 (JP 7048398 B) discloses a technique in which an image showing a plurality of other vehicles is displayed on a display unit.

SUMMARY

[0004] However, in the technique of JP 7048398 B, images showing a plurality of other vehicles are displayed on the display unit in the same manner, and it is not possible to make an occupant of a host vehicle aware of the presence of a specific other vehicle via a display unit.

[0005] It is therefore an object of the present disclosure to provide an information processing device that can make an occupant of a host vehicle aware of the presence of a specific other vehicle via a display unit.

[0006] An information processing device of claim **1** includes a control unit. The control unit is configured to cause a display unit to display an image showing a host vehicle and surroundings of the host vehicle as viewed from a virtual viewpoint, based on surroundings information regarding the surroundings, and is configured to, when a registered vehicle registered in advance is detected, cause the display unit to display an image showing the registered vehicle in a manner different from a manner in which the display unit displays an image showing another vehicle different from the registered vehicle.

[0007] In the information processing device of claim **1**, the control unit causes the display unit to display the image showing the host vehicle and the surroundings as viewed from the virtual viewpoint, based on the surroundings information. When the registered vehicle is detected based on the surroundings information, the control unit causes the display unit to display the image showing the registered vehicle in a manner different from the manner in which the display unit displays the image showing another vehicle different from the registered vehicle. The information processing device can thus make an occupant of the host vehicle aware of the presence of the registered vehicle by causing the display unit to display the image showing the registered vehicle in a manner different from the image showing the other vehicle.

[0008] According to an information processing device of claim **2**, in claim **1**, the control unit may be configured to, when the image showing the registered vehicle becomes unable to be displayed on the display unit based on the surroundings information, erase the image showing the registered vehicle from the display unit and cause the display unit to display location information regarding a current location of the registered vehicle acquired by a communication function to perform communication between the registered vehicle and the host vehicle.

[0009] In the information processing device of claim **2**, when the image showing the registered vehicle becomes unable to be displayed on the display unit based on the surroundings information, the control unit erases the image showing the registered vehicle from the display unit and causes the display unit to display the location information regarding the current location of the registered vehicle acquired by the communication function. The information processing device can thus continue to make the occupant of the host vehicle aware of the current location of the registered vehicle even when the image showing the registered vehicle becomes unable to be displayed on the display unit.

[0010] According to an information processing device of claim **3**, in claim **1** or **2**, the control unit may be configured to, when the image showing the registered vehicle becomes unable to be displayed on the display unit based on the surroundings information, erase the image showing the registered vehicle from the display unit and cause the display unit to display trajectory information regarding a travel trajectory of the registered vehicle acquired by a communication function to perform communication between the registered vehicle and the host vehicle.

[0011] In the information processing device of claim **3**, when the image showing the registered vehicle becomes unable to be displayed on the display unit based on the surroundings information, the control unit erases the image showing the registered vehicle from the display unit and causes the display unit to display the trajectory information regarding the travel trajectory of the registered vehicle acquired by the communication function. The information processing device can thus track the registered vehicle even when the image showing the registered vehicle becomes unable to be displayed on the display unit.

[0012] According to an information processing device of claim **4**, in any one of claims **1** to **3**, the information processing device may further include a contact unit configured to take a predetermined measure to contact an occupant of the registered vehicle, based on an operation on the image showing the registered vehicle that is displayed on the display unit that also serves as an operation unit.

[0013] In the information processing device of claim **4**, the contact unit takes the predetermined measure to contact the occupant of the registered vehicle, based on the operation on the image showing the registered vehicle that is displayed on the display unit that also serves as the operation unit. The information processing device can thus easily contact the occupant of the registered vehicle via the display unit.

[0014] According to an information processing device of claim **5**, in any one of claims **1** to **4**, the information processing device may further include a communication unit configured to send another-vehicle information indicating the other vehicle designated based on an operation on the display unit that also serves as an operation unit to the registered vehicle by a communication function to perform communication between the registered vehicle and the host vehicle.

[0015] In the information processing device of claim **5**, the communication unit sends the other-vehicle information indicating the other vehicle designated based on the operation on the display unit that also serves as the operation unit to the registered vehicle by the communication function. The information processing device can thus make the occupant of the registered vehicle aware of the presence of a specific other vehicle.

[0016] As described above, the information processing device according to the present disclosure can make the occupant of the host vehicle aware of the presence of a specific other vehicle via the display unit.

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0018] FIG. **1** is a block diagram illustrating a hardware configuration of a vehicle;

[0019] FIG. **2** is a flowchart illustrating a flow of a specific process;

[0020] FIG. **3** is a first display example displayed on a monitor;

[0021] FIG. **4** is a second display example displayed on a monitor;

[0022] FIG. **5** is a third exemplary display displayed on a monitor; and

[0023] FIG. **6** is a fourth display example displayed on the monitor.

DETAILED DESCRIPTION OF EMBODIMENTS

[0024] Hereinafter, the vehicle **10** according to the present embodiment will be described.

FIG. **1** is a block diagram illustrating a hardware configuration of a vehicle **10**. As illustrated in FIG. **1**, the vehicles **10** include a meter ECU (Electronic Control Unit) **20**. The vehicle **10** is an example of a "host vehicle", and the meter ECU **20** is an example of an "information processing device".

[0025] The meter ECU **20** includes CPU (Central Processing Unit) **21**, ROM (Read Only Memory) **22**, and RAM (Random Access Memory) **23**. The meter ECU **20** includes a storage **24**, an in-vehicle communication I/F (Interface) **25**, an input/output I/F **26**, and a radio communication I/F **27**. CPU **21**, ROM **22**, RAM **23**, the storage **24**, the in-vehicle communication I/F **25**, the input/output I/F **26**, and the radio communication I/F **27** are communicably connected to each other via an inner bus **28**.

[0026] CPU **21** is a central processing unit that executes various programs and controls each unit. That is, CPU **21** reads the program from ROM **22** or the storage **24**, and executes the program using RAM **23** as a working area. CPU **21** performs control of the above-described configurations and various arithmetic processes in accordance with programs recorded in ROM **22** or the storage **24**.

[0027] ROM **22** stores various programs and various data. RAM **23** temporarily stores a program/data as a working area.

[0028] The storage **24** is constituted by a storage device such as a eMMC (embedded Multi Media Card) or a UFS (Universal Flash Storage), and stores various programs and various data. The storage **24** stores information processing program **24**A. The information processing program **24**A is a program for causing CPU **21** to execute a specifying process (see FIG. **2**), which will be described later.

[0029] The in-vehicle communication I/F **25** is an interface for connecting to another ECU **30**. The interface uses a CAN protocol-based communication standard. The in-vehicle communication I/F **25** is connected to an external bus **29**. Although not shown, a plurality of ECU is provided for each function of the vehicles **10** in addition to ECU **30**.

[0030] The input/output I/F **26** is an interface for communicating with the in-vehicle device **40** mounted on the vehicle **10**.

[0031] The in-vehicle device **40** is a variety of devices mounted on the vehicle **10**. The vehicle **10** includes a sensor group **42** and a monitor **44** as an example of the in-vehicle device **40**.

[0032] The sensor group **42** includes, for example, a 3D-LiDAR, a millimeter-wave sensor, an infra-red sensor, a winker sensor, an accelerator position sensor, a vehicle speed sensor, a steering angle sensor, an angular velocity sensor, and a GPS (Global Positioning System) sensor. The sensor group **42** includes, for example, a sensor for detecting a state of the vehicle **10** and surroundings, such as an illuminance sensor, a gyro sensor, and an acceleration sensor, and a plurality of cameras for capturing an image of the periphery of the vehicle **10**. The sensor group **42** outputs images captured by the respective sensors and images captured by the respective cameras to the meter ECU **20**, ECU **30**, and the like.

[0033] The monitor **44** is a liquid crystal monitor provided on a meter panel disposed in front of a driver's seat of the vehicle **10** and for displaying an operation proposal related to the function of the vehicle **10**, an image related to the explanation of the function, and the like. The monitor **44** integrally includes a touch panel and also serves as an operation unit that can be operated by an occupant of the vehicle **10**. The monitor **44** is an example of a "display unit".

[0034] The wireless communication I/F **27** is a wireless communication module for communicating with an external device. As the radio communication module, for example, communication standards such as 5G, LTE, Wi-Fi (registered trademark) and Bluetooth (registered trademark) are used.

[0035] CPU **21** of the meter ECU **20** includes, as functional components, an acquiring unit **21**A, a control unit **21**B, a communication unit **21**C, and a contact unit **21**D. The respective functional configurations are realized by CPU **21** reading and executing the information processing program **24**A stored in the storage **24**.

[0036] The acquiring unit **21**A acquires various types of data. For example, the acquiring unit **21**A acquires, as various types of information, surroundings information regarding the surroundings of the vehicles **10**. The surroundings information includes a detection result of each sensor constituting the sensor group **42**, an image captured by each camera, and the like.

[0037] The control unit **21**B performs display control related to the display of the monitor **44**. For example, the control unit **21**B causes the monitor **44** to display, as the display control, the host vehicle image **10**A (see FIG. **3** and the like) which is an image showing the vehicle **10** when viewed from the virtual viewpoint and the surroundings image which is an image showing the surroundings of the vehicle **10**, on the basis of the surroundings information acquired by the acquiring unit **21**A. The virtual viewpoint is set on a three-dimensional virtual space whose origin is the position of the host vehicle image **10**A, and is defined by viewpoint coordinates and viewpoint angles (orientations) on the virtual space. For example, the virtual viewpoint is a viewpoint viewed at a specific viewpoint angle from a specific viewpoint coordinate on the upper side of the host vehicle image **10**A in the virtual space.

[0038] Further, the surroundings image includes a road image indicating a road including a roadway and a sidewalk on which the vehicle **10** travels, and a target image indicating a target other than the vehicle **10** such as another vehicle and a pedestrian. The road shown in the road image includes public roads, private roads, empty spaces, square spaces, parking lots, and the like.

[0039] The communication unit **21**C realizes a communication function with other vehicles by using a radio communication I/F **27**. The communication function realizes an inter-vehicle communication system in which the vehicle **10** and other vehicles communicate with each other to directly exchange information on the state and surroundings of each vehicle. For example, according to the inter-vehicle communication system, the vehicle **10** can acquire information on the current location of another vehicle, information on the travel trajectory, and the like. Further, according to the inter-vehicle communication system, the vehicle **10** can notify another vehicle of the presence of a specific other vehicle (e.g., an emergency vehicle).

[0040] The contact unit **21**D takes a predetermined measure to contact an occupant of the registered vehicles registered in advance. An occupant of the vehicle **10** inputs registered vehicle information such as a vehicle name, a vehicle type, a dimension, a shape, a color, an automobile registration number, and contact information of an occupant of the registered vehicle by an input operation using an input interface (not shown), and registers the registered vehicle in advance. The registered vehicle information is stored in the storage **24**. Further, the predetermined measure includes a call, a mail, a message, and the like. In the present embodiment, the contact unit **21**D performs a hands-free call to the telephone number of the occupant of the registered vehicle indicated by the registered vehicle information as the predetermined measure.

[0041] FIG. **2** is a flow chart illustrating a flow of a specifying process executed by the meter ECU **20**. CPU **21** reads the information processing program **24**A from the storage **24**, develops it in RAM **23**, and executes it, thereby performing the specifying process. As an example, the identification processing is performed repeatedly and automatically every time a certain period of time elapses.

[0042] In S**10** illustrated in FIG. **2**, CPU **21** acquires the surroundings information regarding the surroundings of the vehicles **10**. Then, CPU **21** proceeds to S**11**.

[0043] In S**11**, CPU **21** determines whether or not the registered vehicles have been detected based on the surroundings information acquired in S**10**. Here, if it is determined that the registered vehicles have been detected (S**11**: YES), CPU **21** proceeds to S**12**. On the other hand, when CPU **21** determines that the registered vehicles have not been detected (S**11**: NO), the process proceeds to S**13**. As an example, CPU **21** determines that the registered vehicle has been detected when the same number as the vehicle registration number indicated in the registered vehicle information stored in the storage **24** can be recognized from the images captured by the cameras constituting the sensor group **42**.

[0044] In S**12**, CPU **21** displays the registered vehicle image **10**B (see FIG. **3** and the like) which is an image indicating the registered vehicle on the monitor **44** in a particular manner. A particular aspect is a dedicated design configured for a registered vehicle. The specific mode may be set by an occupant of the vehicle **10**, or may be prepared in advance. Then, CPU **21** proceeds to S**13**.

[0045] In S**13**, CPU **21** updates the content displayed on the monitor **44** based on the surroundings information acquired in S**10**. For example, CPU **21** narrows the inter-vehicle distance between the host vehicle image **10**A on the monitor **44** and the registered vehicle image **10**B when the vehicle **10** approaches the registered vehicle ahead. Then, CPU **21** ends the identifying process.

[0046] Next, an exemplary displaying of the host vehicle image **10**A and the surroundings image on the monitor **44** will be described with reference to FIGS. **3** to **6**.

[0047] FIG. **3** is a first explanatory diagram illustrating a display example displayed in the display area X of the monitor **44**. The display area X is a partial area of the monitor **44**, and the occupant seated in the driver's seat can be visually recognized through the opening of the steering wheel.

[0048] In the display area X shown in FIG. **3**, the host vehicle image **10**A and the registered vehicle image **10**B, the other-vehicle image **10**C, and the road image **10**D are displayed as the surroundings image. These images are displayed in the display area X from the viewpoint viewed from the above-described virtual viewpoint. The registered vehicle image **10**B is an image indicating a registered vehicle located forward of the same lane as the vehicle **10**. The other-vehicle image **10**C is an image indicating a different vehicle than the registered vehicle located on the forward side of the right lane of the vehicle **10**. In other words, the other-vehicle image **10**C is an image indicating another vehicle that is not registered as a registered vehicle in the vehicle **10**. The road image **10**D is an image indicating a road on which the vehicles **10** are traveling.

[0049] Here, CPU **21** of the meter ECU **20** displays the host vehicle image **10**A, the registered vehicle image **10**B, and the other-vehicle image **10**C on the road image **10**D based on the acquired surroundings information. For example, as illustrated in FIG. **3**, CPU **21** causes the registered vehicle image **10**B to be displayed on the same lane as the host vehicle image **10**A in the road image **10**D, and causes the other-vehicle image **10**C to be displayed on the right lane of the host vehicle image **10**A in the road image **10**D.

[0050] In addition, CPU **21** assigns a dedicated pattern to the registered vehicle image **10**B as the above-described particular aspect. On the other hand, CPU **21** does not give any pattern to the other-vehicle-image **10**C and expresses it in white. As shown in FIG. **3**, the visibility of the registered vehicle image **10**B is enhanced by displaying the registered vehicle image **10**B on the monitor **44** in a manner that differs from the other-vehicle image **10**C. Note that the above-described different aspects are not limited to changing the pattern of the image, and may be, for example, changing the color of the image or displaying the image in a dedicated shape.

[0051] Further, CPU **21** executes a hands-free call to the occupant of the registered vehicle on the basis of an operation on the registered vehicle image **10**B displayed on the monitor **44**, for example, a touch operation by the finger of the occupant of the vehicle **10**.

[0052] FIG. **4** is a second explanatory diagram illustrating a display example displayed in the display area X of the monitor **44**.

In the display area X illustrated in FIG. **4**, the registered vehicle image **10**B cannot be displayed on the monitor **44** on the basis of the surroundings information by the vehicle **10**, and thus the registered vehicle image **10**B is deleted. When the registered vehicle image **10**B cannot be displayed on the monitor **44** on the basis of the surroundings information, it includes a case where the vehicle **10** cannot detect the registered vehicle from the surroundings information, a case where the location of the registered vehicle detected based on the surroundings information is out of the display area of the monitor **44**, and the like.

[0053] Here, the vehicle **10** and the registered vehicle perform vehicle-to-vehicle communication using a communication function at regular intervals, and transmit information on the state and surroundings of each vehicle to each other. Thus, in the display area X, the location information **10**E regarding the current location of the registered vehicle acquired by the communication function is displayed in place of the registered vehicle image **10**B. In the location information **10**E, the letter "I am 2 km before" is displayed in the balloon to indicate the current location of the registered vehicles.

[0054] FIG. **5** is a third explanatory diagram illustrating a display example displayed in the display area X of the monitor **44**.

In the display area X illustrated in FIG. **5**, the registered vehicle image **10**B cannot be displayed on the monitor **44** on the basis of the surroundings information by the vehicle **10**, and thus the registered vehicle image **10**B is deleted. As described above, the vehicle **10** and the registered vehicle perform vehicle-to-vehicle communication using the communication function at regular intervals, and transmit information on the state and the surroundings of each vehicle to each other. Thus, in the display area X, instead of the registered vehicle image **10**B, the trajectory information **10**F related to the travel trajectory of the registered vehicle acquired by the communication function is displayed. The trajectory information **10**F indicates a travel trajectory of the registered vehicles by arrows on the road image **10**D. For example, according to the trajectory information **10**F, the occupant of the vehicle **10** can recognize that the registered vehicle has traveled straight by changing the lane from the center lane on which the vehicle **10** is traveling to the left lane.

[0055] FIG. **6** is a fourth explanatory diagram illustrating a display example displayed in the display area X of the monitor **44**.

In the display area X shown in FIG. **6**, other-vehicle images **10**G are additionally displayed in the display area X shown in FIG. **3**. The other-vehicle image **10**G is an image indicating a different vehicle than the registered vehicle located on the forward side of the left lane of the vehicle **10**. In other words, the other-vehicle image **10**G is an image indicating another vehicle that is not registered as a registered vehicle in the vehicle **10**.

[0056] Here, a pattern that differs from the registered vehicle image **10**B is given to the other-vehicle image **10**G illustrated in FIG. **6**. At this time, CPU **21** assigns the pattern shown in FIG. **6** to the other-vehicle image **10**G which has been expressed in white based on an operation on the other-vehicle image **10**G displayed on the monitor **44**, for example, a touch operation by a finger of an occupant of the vehicle **10**. Then, CPU **21** transmits the other-vehicle information indicating the other vehicle designated by the other-vehicle image **10**G designated based on the touching operation to the registered vehicle by the communication function. The other-vehicle information includes various types of information such as a vehicle name, a vehicle type, a dimension, a shape, a color, and a vehicle registration number of the other vehicle. At this time, the control device that controls the in-vehicle device of the registered vehicle may change the display mode of the image indicating the other vehicle specified by the other-vehicle information based on the acquisition of the other-vehicle information. For example, the control device may assign a pattern similar to the other vehicle image **10**G shown in FIG. **6** to an image showing the other vehicle being displayed on the monitor of the registered vehicle. In addition, the control device may output, based on the acquisition of the other-vehicle information, the fact that the specific other vehicle indicated in the other-vehicle information is approaching from the speaker as the in-vehicle device by voice. As described above, according to the inter-vehicle communication system of the present embodiment, the occupant of the vehicle **10** can notify the occupant of the registered vehicle of the presence of a specific other vehicle (e.g., an emergency vehicle).

[0057] As described above, in the meter ECU **20**, CPU **21** displays the host vehicle image **10**A and the surroundings image viewed from the virtual viewpoint on the monitor **44** based on the surroundings information. When a registered vehicle is detected based on the surroundings information, CPU **21** displays the registered vehicle image **10**B on the monitor **44** in a manner different from an image indicating another vehicle different from the registered vehicle. Thus, according to the meter ECU **20**, the presence of the registered vehicle can be recognized by the occupant of the vehicle **10** on the monitor **44** by displaying the registered vehicle image **10**B on the monitor **44** differently from the image indicating the other vehicle.

[0058] Further, in the meter ECU **20**, when the registered vehicle image **10**B cannot be displayed on the monitor **44** based on the surroundings information, CPU **21** deletes the registered vehicle image **10**B from the monitor **44** and causes the monitor **44** to display location information **10**E regarding the current location of the registered vehicle acquired by the communication function. Thus, according to the meter ECU **20**, even if the registered vehicle image **10**B cannot be displayed on the monitor **44**, the occupant of the vehicle **10** can continue to grasp the current location of the registered vehicle on the monitor **44**.

[0059] Further, in the meter ECU **20**, when the registered vehicle image **10**B cannot be displayed on the monitor **44** based on the surroundings information, CPU **21** deletes the registered vehicle image **10**B from the monitor **44** and causes the monitor **44** to display the trajectory information **10**F regarding the travel trajectory of the registered vehicle acquired by the communication function. Thus, according to the meter ECU **20**, even if the registered vehicle image **10**B cannot be displayed on the monitor **44**, the registered vehicle can be tracked.

[0060] Further, in the meter ECU **20**, CPU **21** executes a hands-free call to the occupant of the registered vehicle based on the manipulation on the registered vehicle image **10**B displayed on the monitor **44**. Thus, according to the meter ECU **20**, it is possible to easily contact the occupant of the registered vehicles from the monitor **44**.

[0061] Further, in the meter ECU **20**, CPU **21** transmits another-vehicle information indicating the other vehicle designated based on the manipulation of the monitor **44** to the registered vehicle by the communication function. Thus, according to the meter ECU **20**, the occupant of the registered vehicle can recognize the presence of a specific other vehicle.

Others

[0062] In the above-described embodiment, the registered vehicle may be the same another vehicle as the destination of the vehicle **10** as the destination acquired by the inter-vehicle communication, in addition to matching the various types of information indicated in the registered vehicle information stored in the storage **24**. In this case, the vehicle **10** may be configured to track the registered vehicle as a navigation route. With this configuration, the occupant of the vehicle **10** can track the registered vehicle on the basis of the information on the travel trajectory of the registered vehicle acquired by the inter-vehicle communication, even if the registered vehicle being tracked disappears due to a left-right turn or the like at the intersection and cannot be detected from the surroundings information of the vehicle **10**.

[0063] In the above-described embodiment, the vehicle **10** and the registered vehicle perform inter-vehicle communication at regular intervals, but the present disclosure is not limited thereto, and inter-vehicle communication may not be performed. In this case, CPU **21** of the meter ECU **20** continues to display the registered vehicle image **10**B on the monitor **44** while the registered vehicle image **10**B can be displayed on the monitor **44** based on the surroundings information, and deletes the registered vehicle image **10**B from the monitor **44** when the registered vehicle image **10**B cannot be displayed on the monitor **44** based on the surroundings information. Then, unlike the above-described embodiment, CPU **21** does not display the location information **10**E or the trajectory information **10**F on the monitor **44** instead of the registered vehicle image **10**B deleted from the monitor **44**.

[0064] In the above-described embodiment, CPU **21** of the meter ECU **20** is displayed on the monitor **44** using only the target object present in the forward direction of the traveling direction of the vehicles **10** as the target object images, but the present disclosure is not limited thereto. For example, CPU **21** may display, on the monitor **44**, an image indicating another vehicle that is located behind and on the side of the vehicle **10**, as a target image, based on the surroundings information.

[0065] In the above embodiment, the registered vehicle image **10**B is an image indicating another vehicle traveling in the same direction as the vehicle **10**, but the present disclosure is not limited thereto. The registered vehicle image **10**B may be an image indicating another vehicle traveling in a direction other than the vehicle **10**, for example, another vehicle traveling in an oncoming lane. In this case, CPU **21** of the meter ECU **20** adds the patterns shown in FIGS. **3** and **6** to the images showing the registered vehicle represented by the white space being displayed on the monitor **44**, based on the fact that the registered vehicle is detected from the surroundings information. Thus, the occupant of the vehicle **10** can recognize that the vehicle has passed the registered vehicle on the basis of the change in the display mode of the image indicating the registered vehicle.

[0066] In the above-described embodiment, the monitor **44** is an example of the "display unit", but the example of the "display unit" is not limited thereto. For example, the "display unit" may be a multimedia display provided above the center console, or may be a head-up display (HUD). When an example of the "display unit" is another display that differs from the monitor **44**, an example of the information processing device is an ECU that controls the other display.

[0067] In the above-described embodiment, the meter ECU **20** executes the specifying process illustrated in FIG. **2**. However, the present disclosure is not limited thereto, and the specifying process may be executed by the meter ECU **20** and other ECU in cooperation with each other. For example, the meter ECU **20** may execute other ECU to acquire the surroundings information in S**10** and to detect and determine the registered vehicles in S**11**, and control of displaying the monitor **44** in S**12** and S**13**.

[0068] It should be noted that the identifying process executed by CPU **21** reading the software (program) in the above-described embodiment may be executed by various processors other than CPU. Examples of the processor include a PLD (Programmable Logic Device) that can change the circuitry after manufacturing such as FPGA (Field-Programmable Gate Array). Examples of the processor include a dedicated electric circuit or the like, which is a processor having a circuit configuration designed exclusively for executing a particular process such as ASIC (Application Specific Integrated Circuit). Further, the specifying process may be executed by one of these various processors, or may be executed by a combination of two or more processors (for example, a plurality of FPGA, a combination of CPU and FPGA, and the like) of the same type or different types. Further, a hardware structure of the various processors is, more specifically, an electric circuit in which circuit elements such as semiconductor elements are combined.

[0069] In the above-described embodiment, the information processing program **24**A is stored (installed) in the storage **24** in advance, but the present disclosure is not limited thereto. The information processing program **24**A may be provided in a form recorded in a recording medium of CD-ROM (Compact Disk Read Only Memory) or DVD-ROM (Digital Versatile Disk Read Only Memory). The information processing program **24**A may be provided in a form recorded in a recording medium such as a USB (Universal Serial Bus). Further, the information processing program **24**A may be downloaded from an external device via a network.

What is claimed is:

1. An information processing device comprising a control unit configured to cause a display unit to display an image showing a host vehicle and surroundings of the host vehicle as viewed from a virtual viewpoint, based on surroundings information regarding the surroundings, and configured to, when a registered vehicle registered in advance is detected, cause the display unit to display an image showing the registered vehicle in a manner different from a manner in which the display unit displays an image showing another vehicle different from the registered vehicle.

2. The information processing device according to claim **1**, wherein the control unit is configured to, when the image showing the registered vehicle becomes unable to be displayed on the display unit based on the surroundings information, erase the image showing the registered vehicle from the display unit and cause the display unit to display location information regarding a current location of the registered vehicle acquired by a communication function to perform communication between the registered vehicle and the host vehicle.

3. The information processing device according to claim **1**, wherein the control unit is configured to, when the image showing the registered vehicle becomes unable to be displayed on the display unit based on the surroundings information, erase the image showing the registered vehicle from the display unit and cause the display unit to display trajectory information regarding a travel trajectory of the registered vehicle acquired by a communication function to perform communication between the registered vehicle and the host vehicle.

4. The information processing device according to claim **1**, further comprising a contact unit configured to take a predetermined measure to contact an occupant of the registered vehicle, based on an operation on the image showing the registered vehicle that is displayed on the display unit that also serves as an operation unit.

5. The information processing device according to claim **1**, further comprising a communication unit configured to send another-vehicle information indicating the other vehicle designated based on an operation on the display unit that also serves as an operation unit to the registered vehicle by a communication function to perform communication between the registered vehicle and the host vehicle.

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